

Anna Nariyeva

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EDUCATION

Georgia Institute of Technology <i>B.S. in Mechanical Engineering, Concentration in Manufacturing — GPA: 3.50/4.00</i>	May. 2026 – Exp. May. 2028 Atlanta, GA
The Pennsylvania State University <i>B.S. in Mechanical Engineering, Minor in Mechatronics — GPA: 3.70/4.00</i>	Jan. 2024 – May. 2026 Abington, PA
Yonsei University — Korean Language Institute <i>Intensive Korean Course - Intermediate Level</i>	Aug. 2022 – Nov. 2023 Seoul, South Korea

ENGINEERING RESEARCH

Part-time Undergraduate Biomechanical Research Assistant <i>The Pennsylvania State University</i>	Jan. 2025 – May. 2026 Abington, PA
- Performed mechanical testing and data analysis of 100+ 3D-printed polypropylene orthotic braces using Tinius Olsen (tensile testing) equipment to quantify tensile, compressive, and torsional performance parameters on behalf of Korthotics, an Australian orthotics company. - Analyzed experimental data to investigate the effects of print orientation, material type, extrusion temperature, and layer height on the strength and flexibility of 3D-printed components for ankle-foot orthoses (AFOs). - Applied finite element analysis (FEA) in Fusion 360 and compared simulated and experimental results, calibrated material models, and improved the accuracy of digital simulations of additively-manufactured components. - Presented findings at the ASB East Regional Conference at the University Delaware and collaborated with the biomechanical department to develop production methodologies that minimize the cost of manufacturing.	
Full-time Multi-campus REU - Biomechanical Research Assistant <i>The Pennsylvania State University</i>	May. 2025 – Aug. 2025 University Park and Abington, PA
- Quantified the flexural strength and stiffness of 20+ conventionally and commercially fabricated 3D-printed ankle-foot orthoses (AFOs) to establish performance benchmarks for clinically used devices. - Eliminated testing inaccuracy by designing testing protocols and custom fixtures compatible with Tinius Olsen equipment to measure strength and deformation. - Performed finite element analysis (FEA) to evaluate AFO designs and investigate design solutions that meet specified strength and performance requirements. - Worked closely with Dr. Mukul Talaty through weekly research meetings to interpret experimental findings, refine testing methodologies, and develop independent problem-solving skills.	

EXPERIENCE

Part-time Engineering Laboratory Assistant <i>The Pennsylvania State University</i>	Sep. 2025 – May. 2026 Abington, PA
- Assembled, calibrated, and maintained 10+ 3D printers and mechanical testing equipment to support undergraduate engineering coursework and laboratory experiments. - Trained and supervised 30+ undergraduate students in the safe operation of 3D printers and mechanical testing equipment, promoting laboratory safety and reliable experimental procedures. - Troubleshooted laboratory hardware and engineering software, including Prusa and Raise3D printers, SolidWorks, and Fusion 360, while assisting with experimental documentation.	

CONFERENCES

ASB East Regional Conference, University of Delaware <i>Mechanical Performance of Polypropylene for FDM-Based 3D-Printed Ankle-Foot Orthoses</i>	April, 2026
Multi-Campus REU, Penn State University Park <i>Comparing the Mechanical Properties for Conventionally Fabricated and 3D-Printed Ankle-Foot Orthoses</i>	July, 2025

AWARDS

Multi-campus REU Summer Program Scholarship

SKILLS

Engineering: CAD, Finite Element Analysis (FEA), Mechanical Testing, Additive Manufacturing, C++
Software: SolidWorks, Fusion 360, MATLAB, NI Multisim, Quartus Prime, Tinius Olsen Horizon
Manufacturing & Prototyping: FDM printing, SLS printing, FGF printing, PrusaSlicer, Cura
Languages: English (Fluent), Spanish (Native), Russian (Native), Korean (Intermediate)